

OMFS 553

Radiation Biology

Dr.Manila

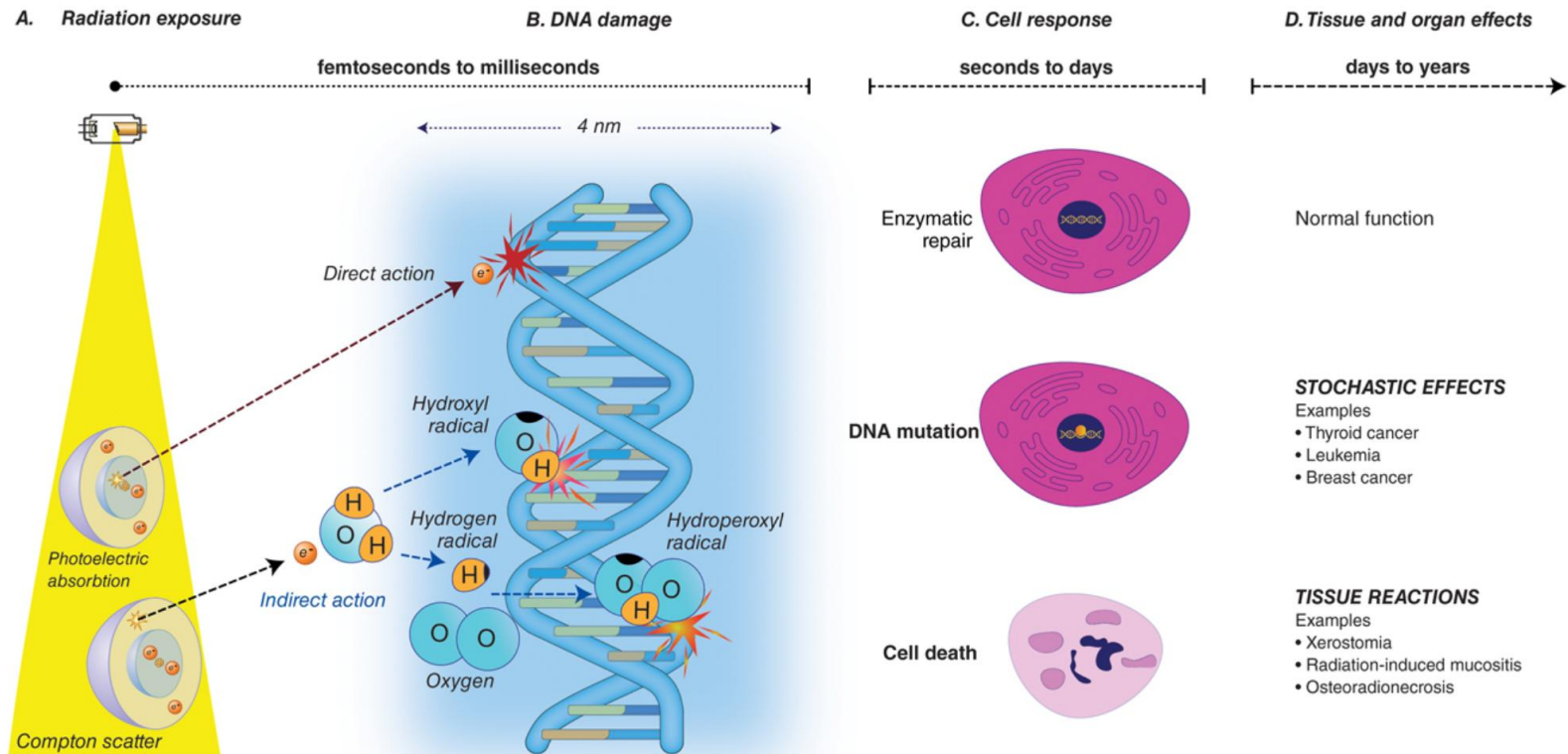
The Biologic Effects of Ionizing Radiation

Radiobiology is the study of the effects of ionizing radiation on living systems

The Biologic Effects of Ionizing Radiation

- ◇ Ionizing radiation has enough energy to remove electrons and create ion pairs.
- ◇ Initial interactions with tissue atoms occur extremely fast ($\approx 10^{-13}$ sec).
- ◇ DNA damage is completed within about 1 millisecond.
- ◇ Cell responses to DNA damage occur from seconds to days.
- ◇ Biological consequences may appear hours, days, or years later depending on damage severity.
- ◇ Foundation for understanding diagnostic & therapeutic radiation effects, especially in oral tissues.

Overview of events after exposure to ionizing radiation.

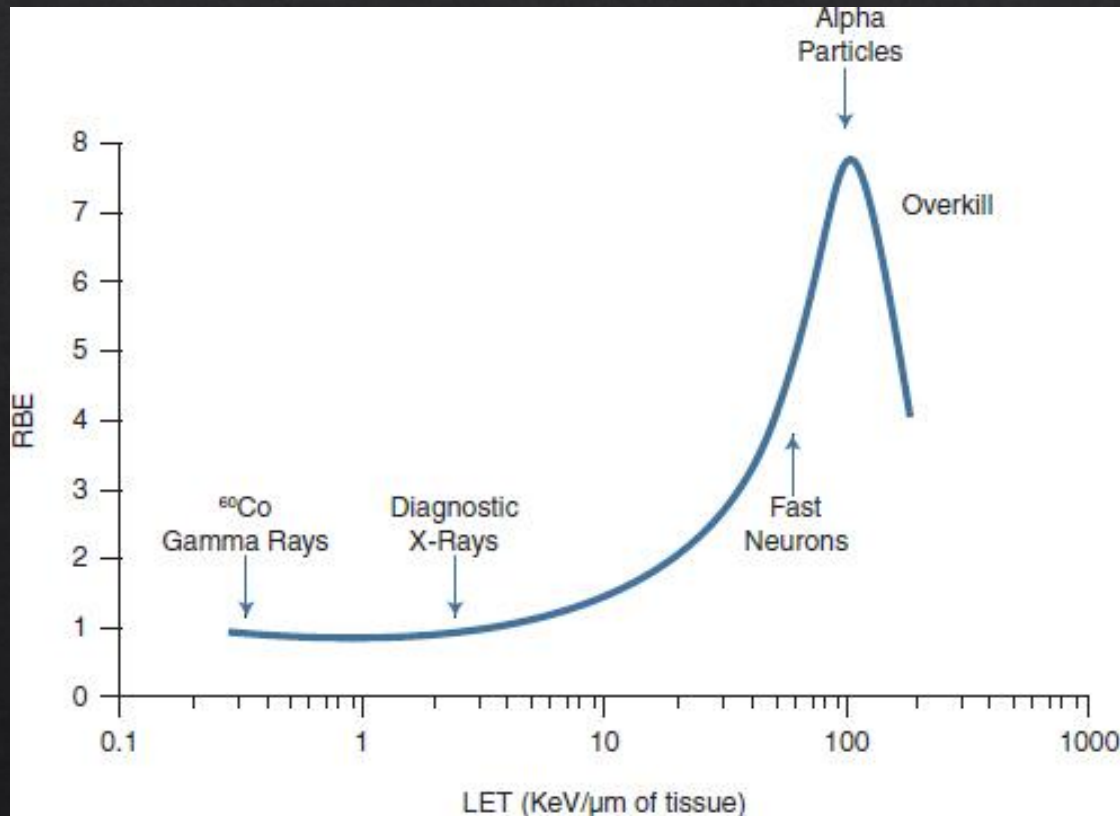


Chemical, molecular, and cellular consequences of radiation absorption

- ◇ Ionizing radiation deposits energy in matter; deposition pattern depends on radiation type.
- ◇ High-velocity charged particles (α , β , electrons, protons) = high LET, densely ionizing, short tissue penetration.
- ◇ X-rays and γ -rays = low LET, less energy deposited per track length, deeper penetration.

Linear energy transfer (LET)

The rate of loss of energy from a particle as it moves along its track through matter (tissue)



Chemical, molecular, and cellular consequences of radiation absorption

◆ Radiographs are made with ionizing radiation

◆ Risk for cell damage from radiation is small but NOT zero

◆ Follow ALARA principle to minimize risks



Chemical, molecular, and cellular consequences of radiation absorption

- ◇ ALARA principle

- ◇ $A = As$

- ◇ $L = \text{low}$

- ◇ $A = \text{as}$

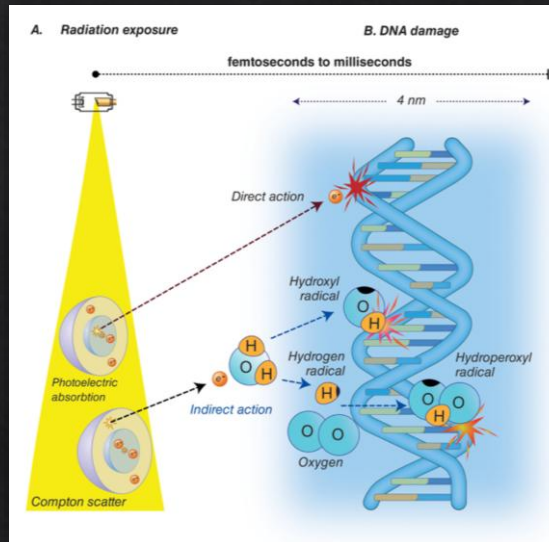
- ◇ $R = \text{reasonably}$

- ◇ $A = \text{achievable}$

- ◇ Avoiding exposure to radiation that does not have a direct benefit to you, even if the dose is small.

Chemical, molecular, and cellular consequences of radiation absorption

Biologic effects of ionizing radiation occur through direct and indirect actions



Direct Action:

- Photons or secondary electrons directly ionize DNA, causing immediate molecular damage.
- Secondary electrons also contribute to direct DNA injury.

Indirect Action:

- Photons/secondary electrons ionize water molecules, producing free radicals.
- Free radicals and oxidizing species diffuse to damage DNA indirectly.

Direct Effect

- ◇ DNA absorbs energy from photons or secondary electrons → ionization of bases or sugar-phosphate backbone.
- ◇ Free radicals form within 10^{-10} seconds and cause dissociation or cross-linking.
- ◇ High-LET radiation → more direct actions; low-LET (x-rays, gamma) → ~30% of DNA damage is direct.

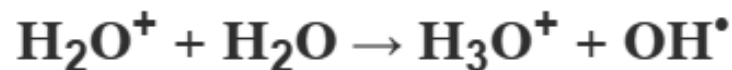


Indirect Effect

- ◇ Indirect effects of x-ray absorption
 - ◇ Water absorbs x-ray energy
 - ◇ Forms variety of free radicals
 - ◇ Unstable compounds that react rapidly
 - ◇ Also forms hydrogen peroxide (toxic)
 - ◇ Free radicals interact with other molecules, causing damage

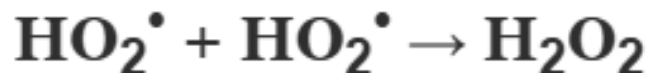
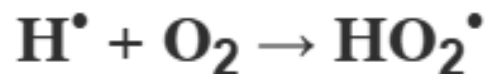
Indirect Effect

- ◇ Photon/secondary electron first interacts with water (~70% of cell mass).
- ◇ Interaction triggers radiolysis of water, creating reactive chemical species.
- ◇ Chain of chemical changes ultimately leads to indirect DNA damage.



Indirect Effect

- ◇ Dissolved oxygen changes the types of free radicals formed during water radiolysis.
- ◇ Oxygen presence leads to formation of ($\text{HO}_2^\bullet$) radicals and hydrogen peroxide.
- ◇ These oxygen-dependent species contribute significantly to DNA damage.



Indirect Effect

- ◇ Radiolysis products of water are highly reactive with short half-lives.
- ◇ Radicals can damage DNA only if created within ~4 nm of the DNA helix.
- ◇ Hydroxyl radicals cause about two-thirds of x-ray-induced biologic damage.

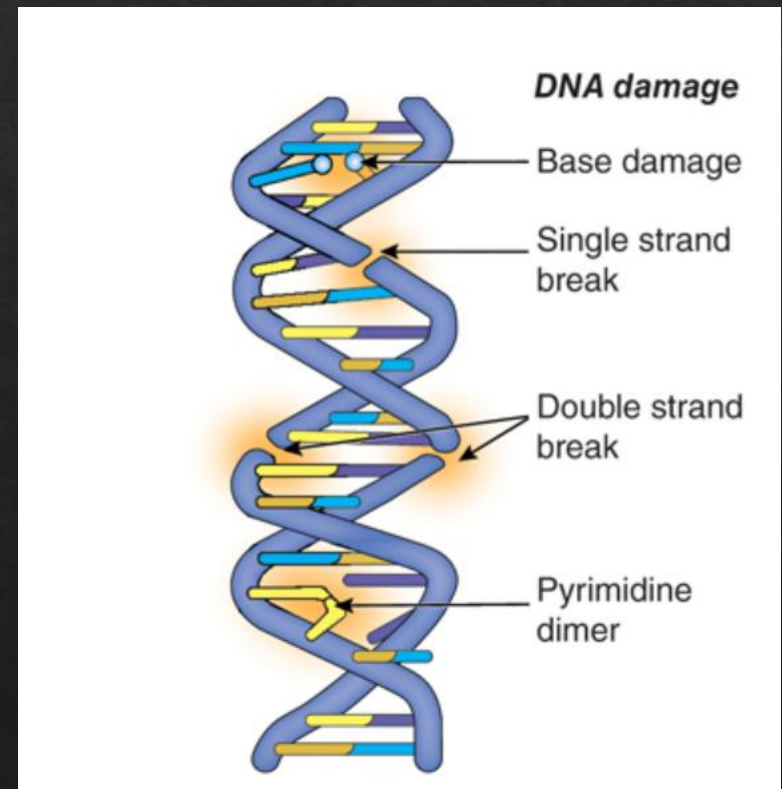


Free Radical

- ◇ Free radical = atom/molecule with an **unpaired electron**, highly reactive.
- ◇ Aggressively binds to **DNA**, causing damage.
- ◇ In presence of **oxygen**, forms hydroxyl radicals-among the most damaging.
- ◇ Primary mediator of the **indirect effects** on DNA

DNA and chromosomal damage and damage response

- ◇ DNA damage is the primary cause of radiation-induced cell death, carcinogenesis, and heritable mutations.
- ◇ Direct & indirect actions cause base damage, single- and double-strand breaks, and DNA/DNA-protein cross-links.

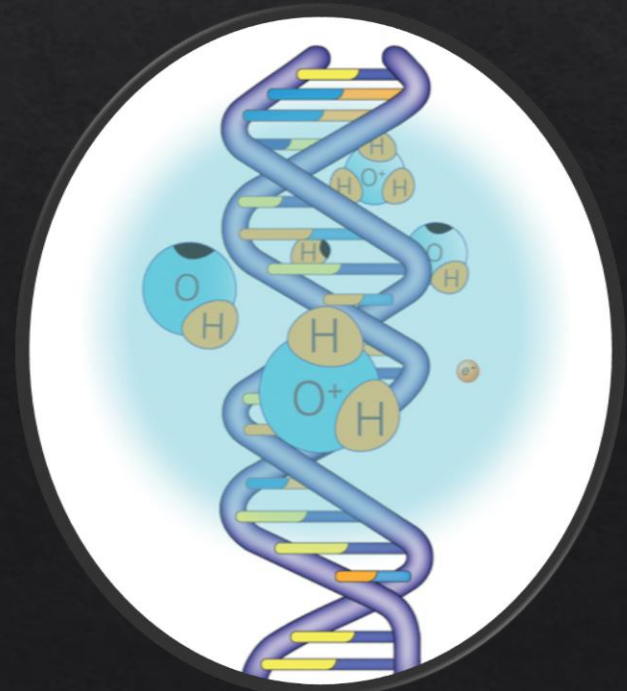
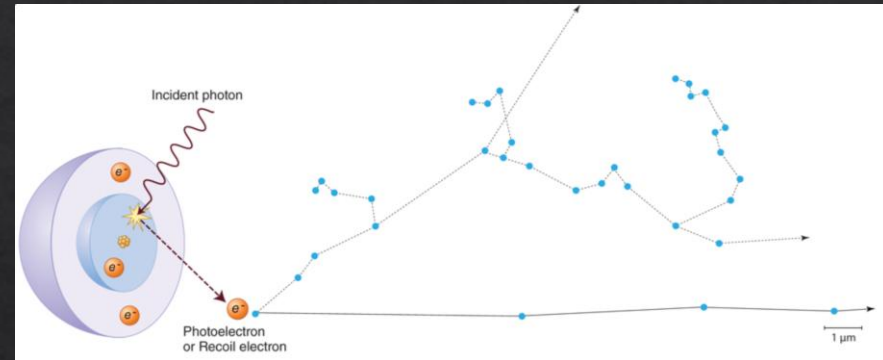


DNA and chromosomal damage and damage response

- ◇ X-ray-induced DNA damage is mostly base damage (72%) and single-strand breaks (27%).
- ◇ Double-strand breaks (1%) are rare but highly significant for cell killing and tumorigenesis.
- ◇ DSBs are 25 times less frequent than SSBs but far more biologically important.

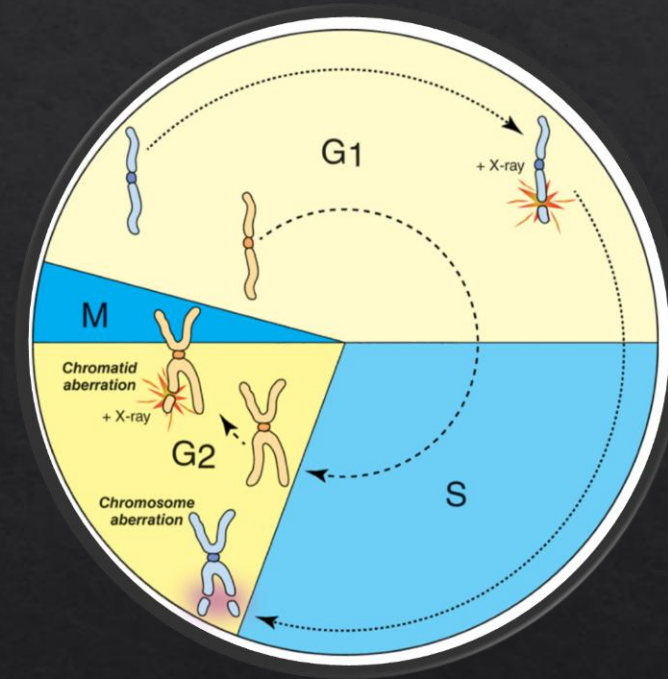
DNA and chromosomal damage and damage response

- ◇ ~95% of x-ray energy is deposited as spurs (~100 eV within 4-nm diameter).
- ◇ Spurs are physically separated; x-rays have a sparsely ionizing track pattern.
- ◇ X-rays produce mostly base damage, with SSBs and DSBs rare.
- ◇ Single spur overlapping the 2-nm DNA helix can create clustered lesions.
- ◇ Clustered lesions occur $\sim 3\times$ more frequently than double-strand breaks.



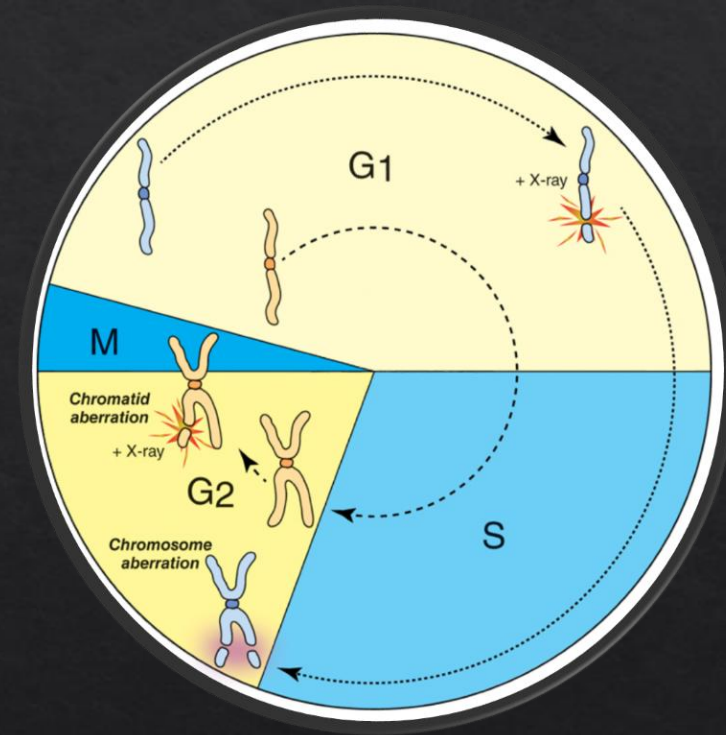
Cell cycle and DNA aberrations

- ◇ G1= gap 1 in which nuclear components are replicated
- ◇ S= synthesis phase; DNA is synthesized during the last 2/3 of this phase
- ◇ G2= gap2 phase, a preparatory stage to cell division
- ◇ M= mitosis during which cells divide



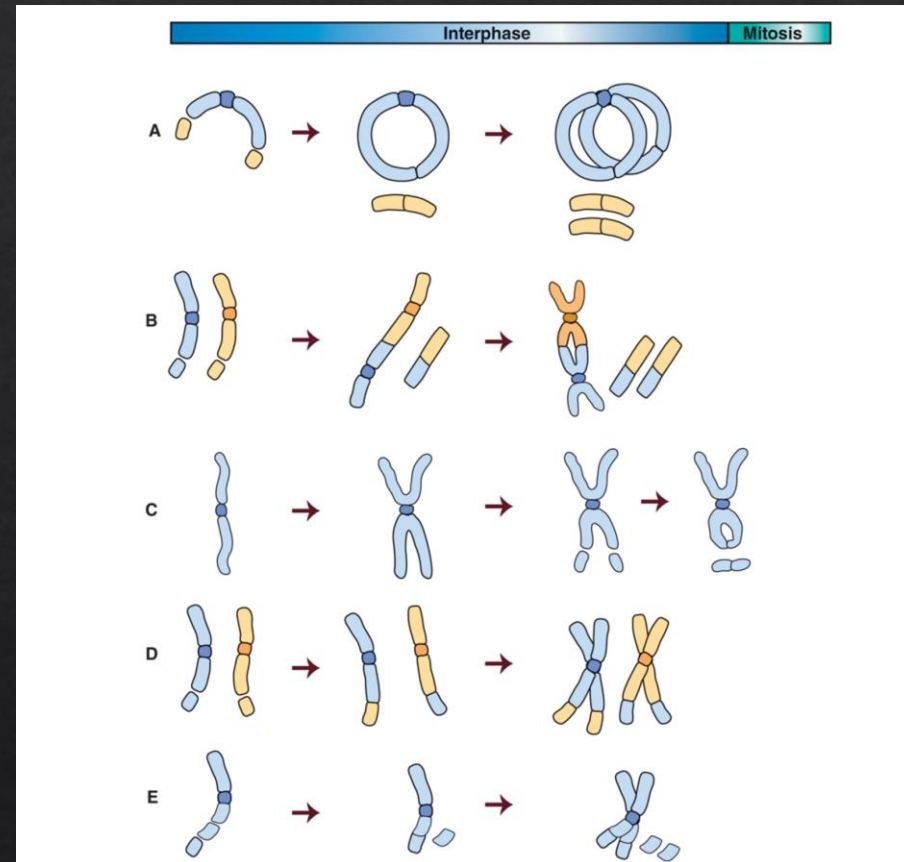
Cell cycle and DNA aberrations

- ◇ Cells cycle through M, G₁, S, and G₂ phases.
- ◇ Most radiosensitive: G₂ and mitosis phases.
- ◇ Less sensitive: G₁ phase.
- ◇ Least sensitive: late S phase.
- ◇ Damage before DNA synthesis → chromosome (double-arm) aberration.
- ◇ Damage after DNA synthesis → chromatid (single-arm) aberration.



Cell cycle and DNA aberrations

- ◇ Chromosome and chromatid aberrations.
 - ◇ (A) Ring formation plus acentric fragment.
 - ◇ (B) Dicentric formation.
 - ◇ (C) Anaphase bridge formation.
 - ◇ (D) Translocation
 - ◇ (E) deletion.
- ◇ The occurrence of the breakage and fusion events relative to the cell cycle phase is denoted by the bar on the top.



Cell killing

Mitotic death

- ◇ The main mechanism of radiation-induced cell killing.
- ◇ Caused by lethal chromosomal and chromatid aberrations.
- ◇ A cell's radiosensitivity depends on its mitotic rate and degree of differentiation.
- ◇ This relationship is known as the Law of Bergonié and Tribondeau.

Cell killing

Casarett's Categorization of Cell Types by Radiosensitivity

Characteristics				
Radiosensitivity	Category	Cell Division	Differentiation Status	Examples
High	I. Vegetative intermitotic	Rapid	Undifferentiated	Basal cells of oral mucosa, bone marrow stem cells
↑	II. Differentiating intermitotic	Regular	Some differentiation	Myelocytes, spermatocytes
Intermediate	III. Multipotent connective tissue			Fibroblasts, endothelial cells
↑	IV. Reverting postmitotic	Not regular but can be stimulated to divide	Fully differentiated	Hepatocytes
Low	V. Fixed postmitotic	No	Highly differentiated	Neurons, muscle cells

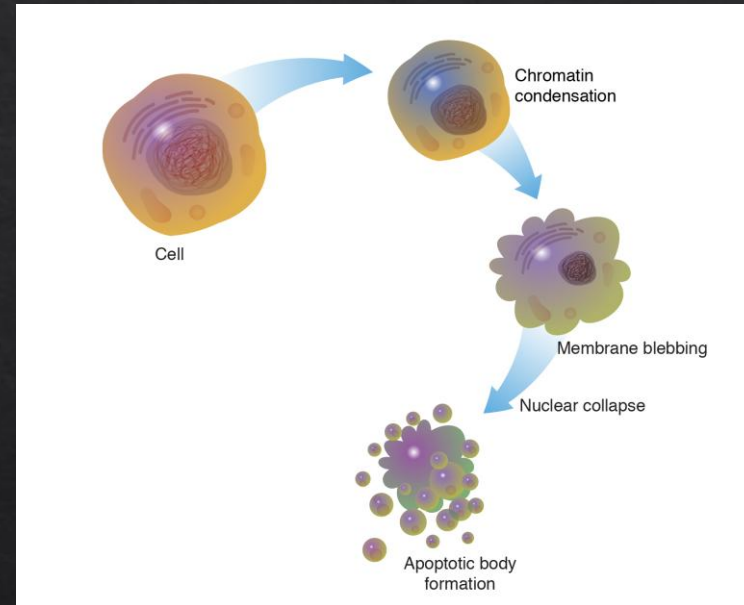
Modified from Rubin P, Casarett GW. *Clinical Radiation Pathology*. Saunders; 1968.

- This rule predicts that Rapidly dividing cells are more radiosensitive.
- Highly specialized, post-mitotic cells are most radioresistant.
- Casarett classification organizes tissues by their radiosensitivity.
- System helps predict which tissues may show radiation-induced reactions.

Cell killing

Apoptosis

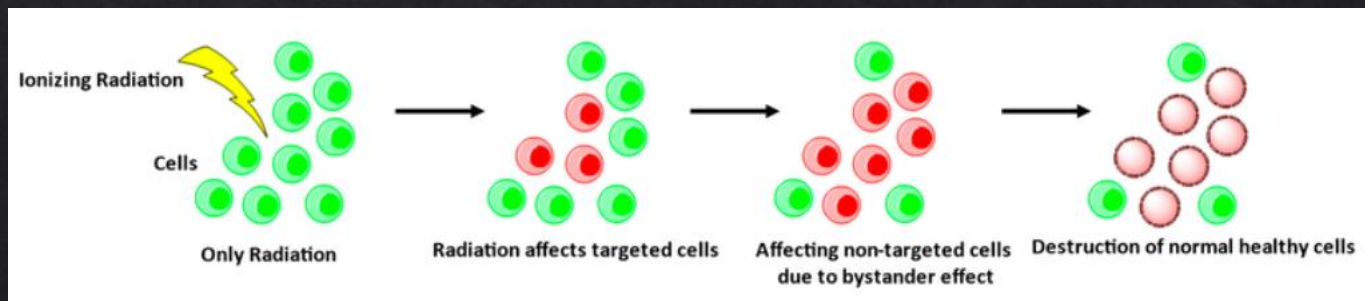
- ◇ Lymphocytes are highly radiosensitive and die mainly by apoptosis.
- ◇ Serous salivary gland cells also undergo apoptotic cell death despite low mitotic activity.
- ◇ Apoptosis occurs within hours after radiation, without requiring mitosis.
- ◇ Triggered by radiation-induced DNA damage.
- ◇ p53 plays a central role in regulating radiation-induced apoptosis.



Cell killing

Bystander effect

- ◆ Nearby non-irradiated cells show radiation-induced responses.
- ◆ Effects include gene expression changes, DNA damage, altered proliferation, and cell death.
- ◆ Mechanism may involve intercellular signaling, chromosome fragments, or diffusible chemokines.



Timeline of radiation damage

- Development of radiation damage consists of following phases:
 - physical phase: $\sim 10^{-12}$ s: ionization
 - chemical phase: $\sim 10^{-6}$ s: formation and diffusion of free radicals
 $\sim 10^{-3}$ s: DNA damage
 - biological phase: $\sim$ seconds: DNA repair
 $\sim$ hours: cell death
 $\sim$ days: mutations, cell transformation,
chromosome aberrations
 $\sim$ years: development of cancer, hereditary
effects



IAEA

International Atomic Energy Agency

Biologic manifestations of ionizing radiation damage

- ◇ Two categories: stochastic effects and tissue reactions.
- ◇ Key difference: stochastic effects have no dose threshold; tissue reactions require a threshold dose.
- ◇ Diagnostic radiation doses are below thresholds for tissue reactions but can still cause stochastic risks.

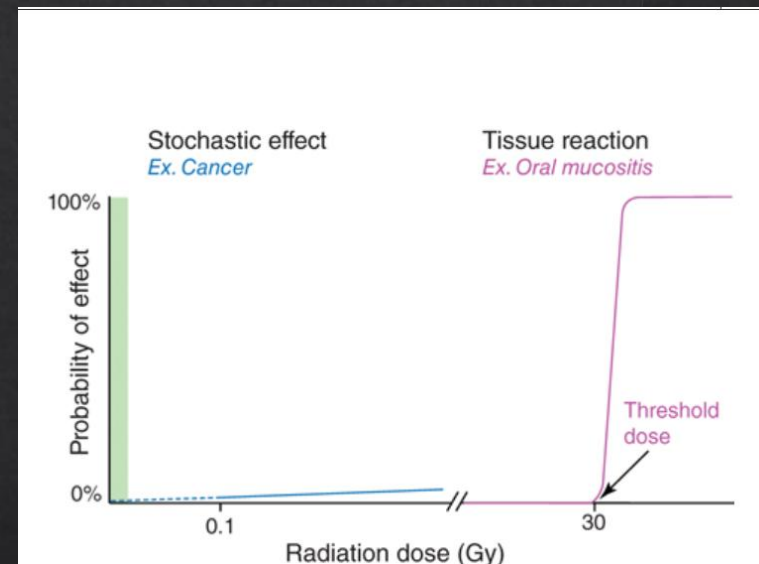
Comparison of Stochastic Effects and Tissue Reactions of Radiation

Stochastic Effects	
Caused by	Sublethal DNA damage
Threshold dose	No
	There is no minimum threshold dose. Effect can be caused by any dose of radiation
Severity of clinical effects and dose	Severity of clinical effects is independent of dose; all-or-none response—an individual either manifests effect or does not
Relationship between dose and effect	Frequency of effect proportional to dose; the higher the dose, the higher the risk of manifesting the effect
Caused by doses used in diagnostic radiology	Yes
Examples	Radiation-induced cancer
	Heritable effects
	Radiation-induced skin cancer

Tissue Reactions	
Caused by	Cell killing
Threshold dose	Yes
	Effect occurs only when the threshold dose is exceeded
Severity of clinical effects and dose	Severity of clinical effects is proportional to dose; the higher the dose, the more severe the effect
Relationship between dose and effect	Probability of effect independent of dose; most individuals manifest effect when threshold dose is exceeded
Caused by doses used in diagnostic radiology	No
Examples	Osteoradionecrosis
	Radiation-induced cataract formation
	Radiation-induced skin burns

Biologic manifestations of ionizing radiation damage

- ◇ Cancer risk increases linearly with doses >0.1 Gy.
- ◇ Stochastic effects have no threshold; risk extends into low-dose range.
- ◇ Tissue reactions (e.g., mucositis) occur only above threshold (≈ 30 Gy).
- ◇ Diagnostic imaging doses lie in a low-risk range.



Stochastic effects

- ◇ Result of sublethal damage to DNA
- ◇ No threshold: any amount of radiation can cause effect
- ◇ Severity of effect independent of the dose
- ◇ Probability of effect proportional to the dose
- ◇ Examples: cancer, heritable effects

General Features of Radiation-Induced Cancer

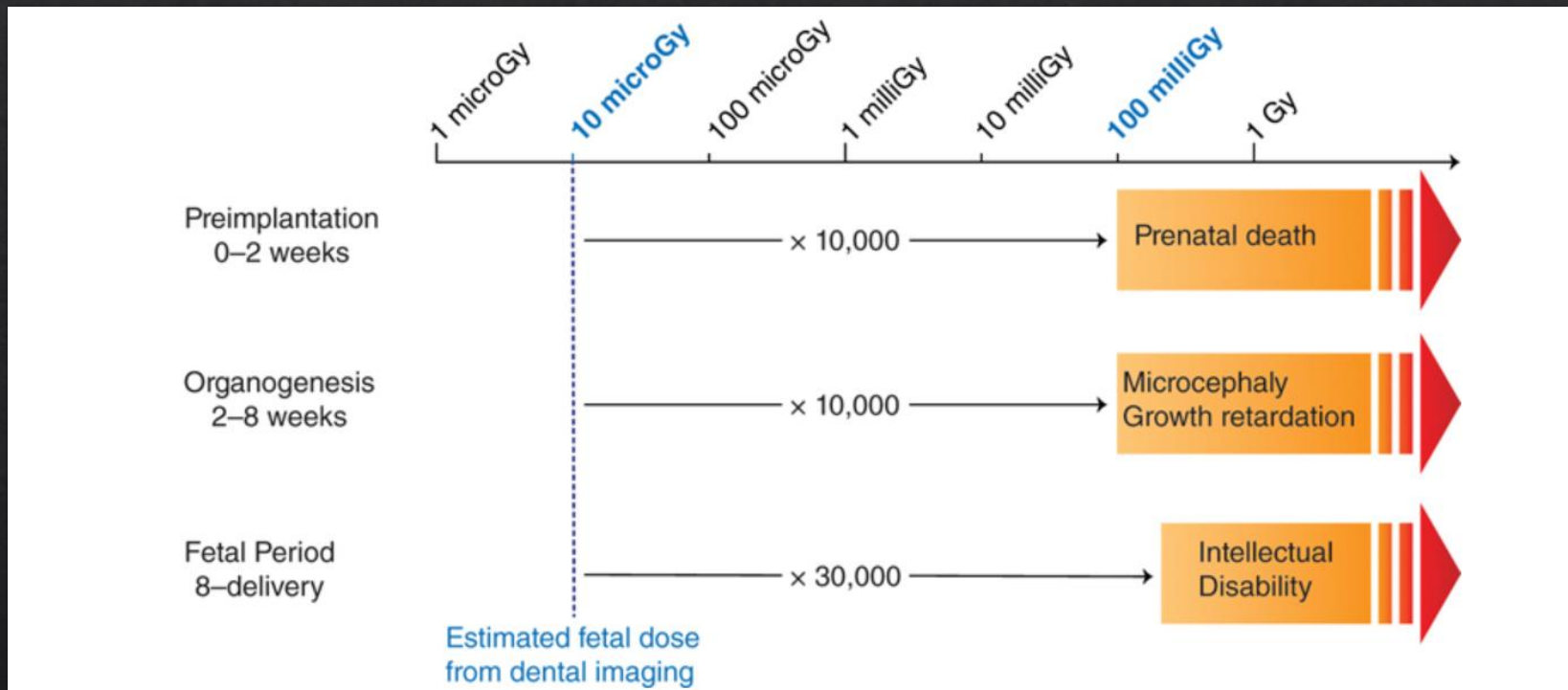
- Radiation-induced cancers are clinically and histologically indistinguishable from sporadically occurring or chemical-induced cancers.
- Certain tissues, for example, the female breast, the thyroid gland, and the red bone marrow, are more sensitive to the carcinogenic effects of ionizing radiation.
- There is a long latent period, ranging from years to decades, between radiation exposure and the occurrence of cancer.
- The risk of radiation-induced tumor induction is approximately threefold higher in children than in adults.

Tissue reactions (deterministic effects)

- ◇ Result from killing large numbers of cells.
- ◇ Have a threshold dose before effects appear.
- ◇ Severity increases with dose; probability does not.
- ◇ Examples: salivary dysfunction, cataracts, hair loss, sterility.
- ◇ Diagnostic exams stay below threshold, so deterministic effects aren't seen.

Tissue reactions (deterministic effects)

Radiation effects on the embryo and fetus



The estimated fetal dose from dentomaxillofacial imaging is indicated (blue dashed line), and its relative magnitude to the threshold doses for effect causation are shown.

Tissue reactions (deterministic effects)

Cataracts

- ◇ Radiation can induce posterior subcapsular or cortical cataracts.
- ◇ Threshold for cataract formation: ~ 0.5 Gy (ICRP); 1–2 Gy (NCRP).
- ◇ Dental radiographs deliver 0.02–0.4 mGy to the lens-far below threshold.

Relative Radiosensitivity of Various Organs

High

Lymphoid
organs
Bone marrow
Testes
Intestines
Mucous
membranes

Intermediate

Fine vasculature
Growing
cartilage
Growing bone
Salivary glands
Lungs
Kidney
Liver

Low

Optic lens
Muscle
Nerves

Effects of therapeutic radiation on oral cavity

Mucositis



Osteoradionecrosis

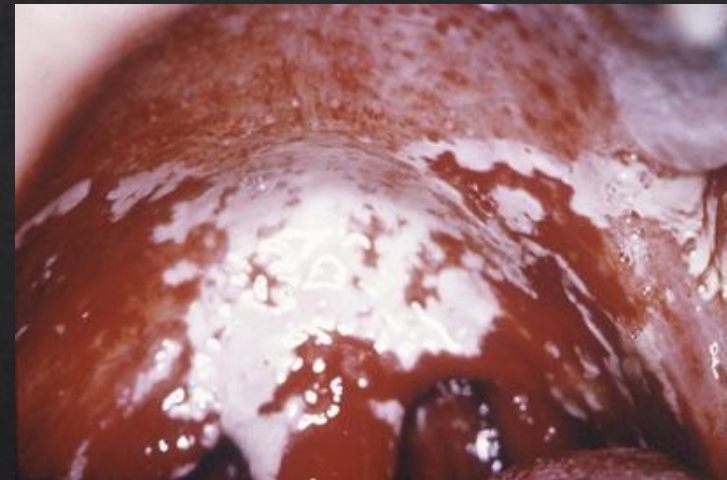


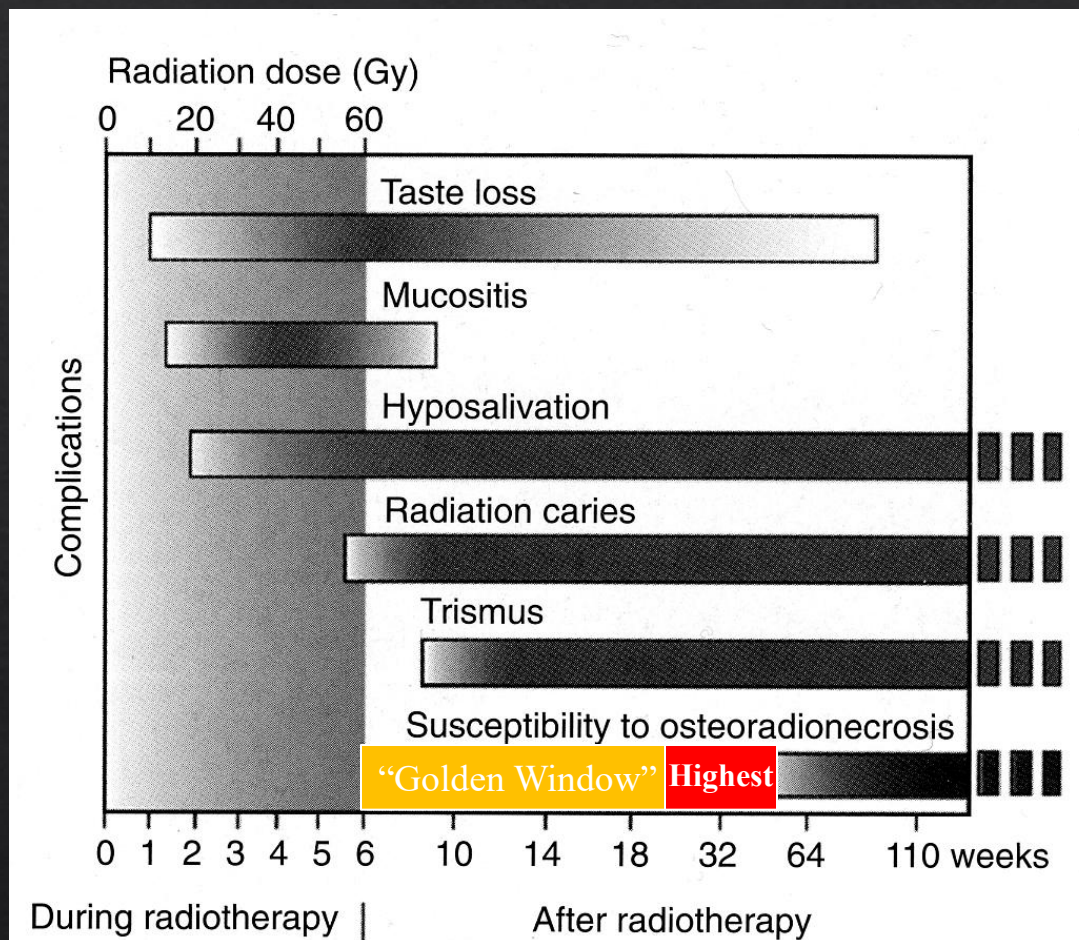
Skin burn & fibrosis

Radiotherapy involving the oral cavity

A- Oral mucous membrane:

1. Redness and inflammation of mm-mucositis
2. White to yellow pseudomembrane
3. At the end of therapy severe discomfort and difficulty to eat
4. Dry infection with candida albicans
5. Healing within 2 months





“Golden Window” = lowest risk of osteoradionecrosis (4 mo after Rx. therapy)
Highest risk: between 4 and 12 months after Rx. therapy

Radiation effect on oral tissue

B-Taste buds:

- ◇ They are sensitive to radiotherapy
- ◇ Patient feels loss of taste during 2nd and 3rd week of radiotherapy
- ◇ Irradiation to posterior 2/3 of tongue affect bitter and acid flavors while anterior 1/3 affect sweet and salt flavors
- ◇ Recovery takes about 60 to 120 days after irradiation

Radiation effect on oral tissue

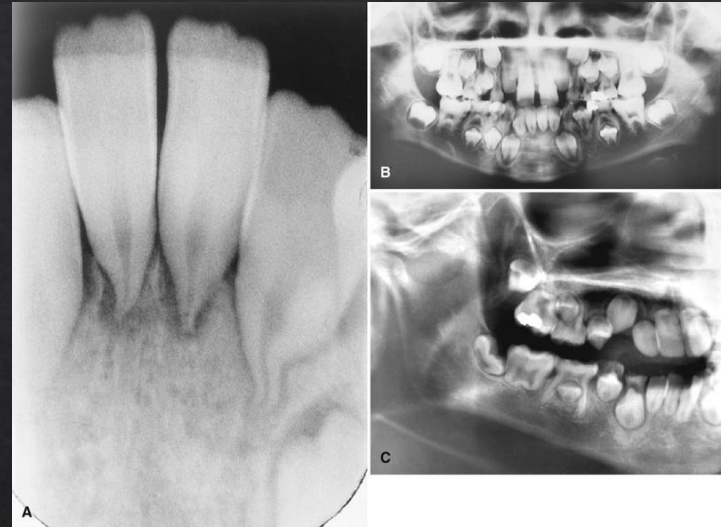
C- Salivary glands

- ◇ The parotid glands are more radiosensitive rather than submand/sublingual
- ◇ Acute inflammatory response soon develop
- ◇ Chronic inflammatory progression leads to progressive fibrosis, adiposis, loss of fine vasculature and degeneration
- ◇ After few weeks of radiotherapy, salivary flow decreases to zero at 60Gy
- ◇ Xerostomia subsides in 6-12 months because of compensatory hypertrophy of residual s.glands. Beyond a year less likely to show significant return of function

Radiation effect on oral tissue

D- Teeth

- ◆ Irradiation of teeth during their development retards their growth
- ◆ If it precedes calcification, it may destroy tooth bud. After calcification it may inhibit cellular differentiation causing malformation and arresting general growth
- ◆ Children receiving radiation therapy shows defects in permanent dentition such as retarded root development or missing 1 or 2



Radiation effect on oral tissue

- ◇ Radiation caries
- ◇ It is a rampant form, result from
 - ◇ Reduced salivary flow
 - ◇ Decreased pH
 - ◇ Reduced buffering capacity
 - ◇ Increased viscosity
- ◇ Clinical presentation
 - ◇ 3-types
 - ◇ Widespread superficial lesions on all buccal, lingual, incisal and palatal surfaces
 - ◇ Circumferential cervically involving dentine and cementum
 - ◇ Dark pigmentation of entire crown and incisor edges may be worn



Radiation effect on oral tissue

- ◊ E- Bone
- ◊ The primary damage of mature bone due to damage of fine vasculature specifically in mandible rather than maxilla
- ◊ The marrow tissue becomes hypovascularized, hypoxic, and hypocellular with replacement of normal marrow with fatty marrow of fibrous connective tissue

Radiation effect on oral tissue

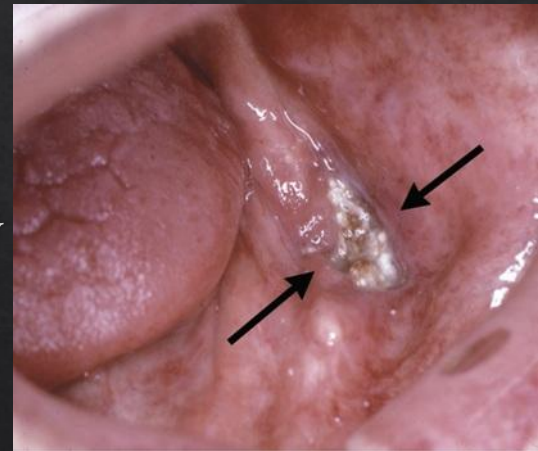
Osteoradionecrosis (ORN)

- ◇ ORN usually appears 6–12 months after radiation but may occur years later.
- ◇ Incidence is ~5–7% with modern radiotherapy (IMRT, brachytherapy).
- ◇ More common in the mandible; rare below 50 Gy, risk increases 11-fold above 66 Gy.
- ◇ Major risk factors: caries, periodontal disease, dental extractions, ill-fitting dentures.
- ◇ Early sign: widened PDL space; advanced disease shows patchy radiolucency, sequestra, possible fracture.

Radiation effect on oral tissue

E- Bone

- ◊ ORN = non-healing wound from mucosal breakdown leading to bone infection and necrosis.
- ◊ Extractions recommended 2-3 weeks before radiation.
- ◊ Post-radiation extractions should include alvelectomy to reduce trauma.
- ◊ Hyperbaric oxygen preferred for post-irradiation extraction management.



Postradiation therapy patient



Image acquired immediately after radiation therapy, patient shows presence of mild periodontal bone loss



Same patient taken 3 years later shows extensive bone destruction in the left posterior mandible body-angle region, extending to the inferior mandibular cortex